Mohamed Sameh Elkholy
Apache – Lab 1

Q1:

```
Activities   Terminal                    Mar 8  2:53 PM                    en   36 %
mohamed@Mohamed-Sameh-PC: ~
mohamed@Mohamed-Sameh-PC:~$ sudo apt update
[sudo] password for mohamed:
Hit:1 http://eg.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://eg.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://eg.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:5 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:6 https://packages.microsoft.com/repos/code stable InRelease
Hit:7 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/8.0 InRelease
Hit:8 https://debrepo.freedownloadmanager.org jammy InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
24 packages can be upgraded. Run 'apt list --upgradable' to see them.
mohamed@Mohamed-Sameh-PC:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1
  libaprutil1 libaprutil1-dbd-sqlite3
  libaprutil1-ldap
Suggested packages:
  apache2-doc apache2-suexec-pristine
  | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils
  libapr1 libaprutil1 libaprutil1-dbd-sqlite3
  libaprutil1-ldap
0 upgraded, 8 newly installed, 0 to remove and 24 not upgraded.
Need to get 1,933 kB of archives.
After this operation, 7,756 kB of additional disk space will be used.
Get:1 http://eg.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://eg.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://eg.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://eg.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9,170 B]
Get:5 http://eg.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.18 [1,360 kB]
Get:6 http://eg.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.18 [165 kB]
```

```
Activities   Terminal                    Mar 8  2:53 PM                    en   36 %
mohamed@Mohamed-Sameh-PC: ~
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.w
ants/apache2.service → /lib/systemd/system/apache2.serv
ice.
Created symlink /etc/systemd/system/multi-user.target.w
ants/apache-htcacheclean.service → /lib/systemd/system/
apache-htcacheclean.service.
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
mohamed@Mohamed-Sameh-PC:~$ sudo systemctl start apache2
mohamed@Mohamed-Sameh-PC:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable apache2
mohamed@Mohamed-Sameh-PC:~$
```

Q2:

```
mohamed@Mohamed-Sameh-PC:~$ sudo nano /var/www/html/page1.html
mohamed@Mohamed-Sameh-PC:~$ sudo nano /var/www/html/page2.html
mohamed@Mohamed-Sameh-PC:~$
```

Q3:

```
mohamed@Mohamed-Sameh-PC:~$ sudo nano /etc/apache2/sites-available/000-default.conf
mohamed@Mohamed-Sameh-PC:~$ sudo systemctl reload apache2
mohamed@Mohamed-Sameh-PC:~$
```

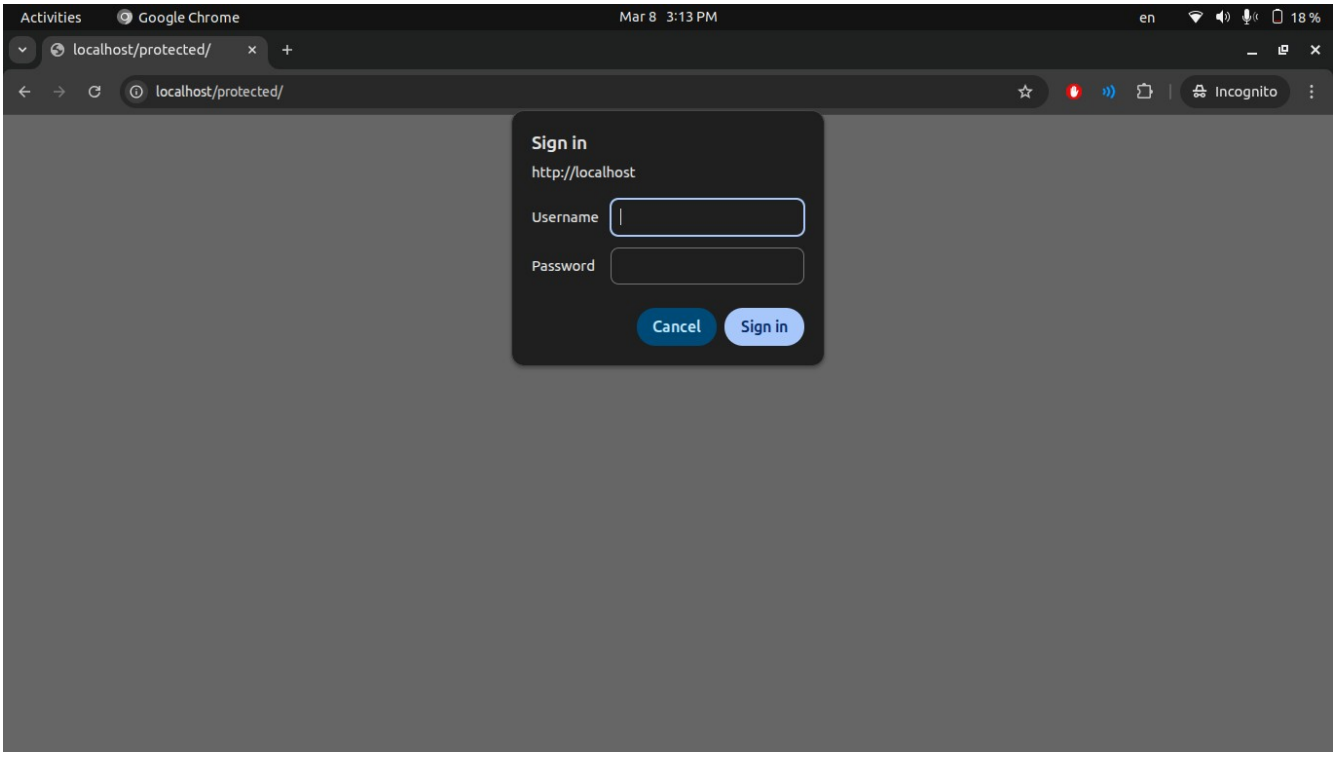
```
Activities   Terminal                    Mar 8  3:04 PM                    en   25 %
mohamed@Mohamed-Sameh-PC: ~
  GNU nano 6.2                          /etc/apache2/sites-available/000-default.conf
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        Redirect /page1.html http://localhost/page2.html

        ServerAdmin webmaster@localhost
        DocumentRoot /var/www/html

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet

                                                [ Read 33 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^Q Where Was
```

Q4:

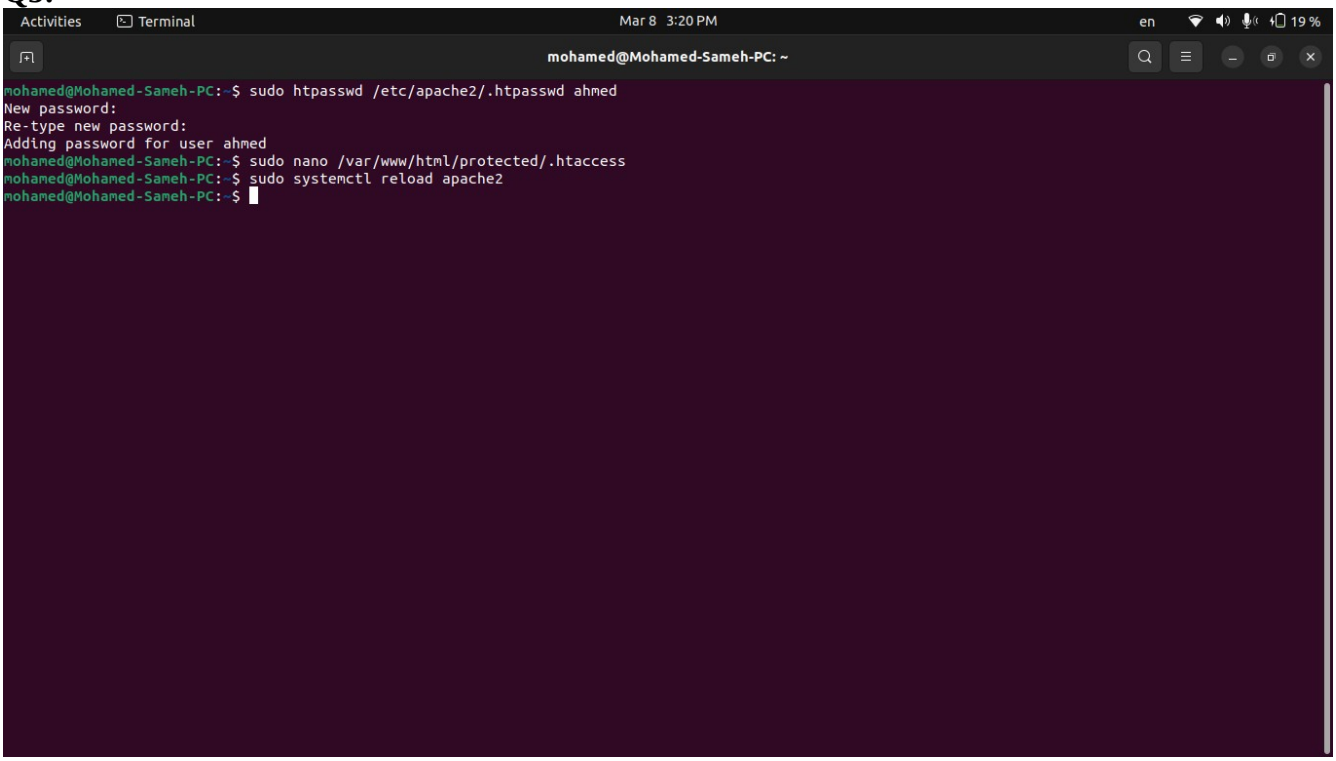
```
Activities   Terminal                    Mar 8  3:11 PM                    en   19 %
mohamed@Mohamed-Sameh-PC: ~
mohamed@Mohamed-Sameh-PC:~$ sudo mkdir /var/www/html/protected
mohamed@Mohamed-Sameh-PC:~$ echo "<h1>Secret Area</h1>" | sudo tee /var/www/html/protected/index.html
<h1>Secret Area</h1>
mohamed@Mohamed-Sameh-PC:~$ sudo htpasswd -c /etc/apache2/.htpasswd student
New password:
Re-type new password:
Adding password for user student
mohamed@Mohamed-Sameh-PC:~$ sudo nano /var/www/html/protected/.htaccess
mohamed@Mohamed-Sameh-PC:~$ sudo nano /etc/apache2/sites-available/000-default.conf
mohamed@Mohamed-Sameh-PC:~$ sudo systemctl reload apache2
mohamed@Mohamed-Sameh-PC:~$
```

```
Activities   Terminal                    Mar 8  3:18 PM                    en   18 %
mohamed@Mohamed-Sameh-PC: ~
  GNU nano 6.2                          /var/www/html/protected/.htaccess
AuthType Basic
AuthName "Restricted Area"
AuthUserFile /etc/apache2/.htpasswd
Require valid-user

                                        [ Read 4 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^Q Where Was
```

```
Activities   Terminal                    Mar 8  3:11 PM                    en   19 %
mohamed@Mohamed-Sameh-PC: ~
  GNU nano 6.2                          /etc/apache2/sites-available/000-default.conf *
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        Redirect /page1.html http://localhost/page2.html

        <Directory /var/www/html/protected>
                AllowOverride All
        </Directory>

        ServerAdmin webmaster@localhost
        DocumentRoot /var/www/html

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^Q Where Was
```


Activities
Google Chrome
Mar 8 3:13 PM
en
18 %
localhost/protected/
localhost/protected/
Incognito
Sign in
http://localhost
Username
Password
Cancel
Sign in

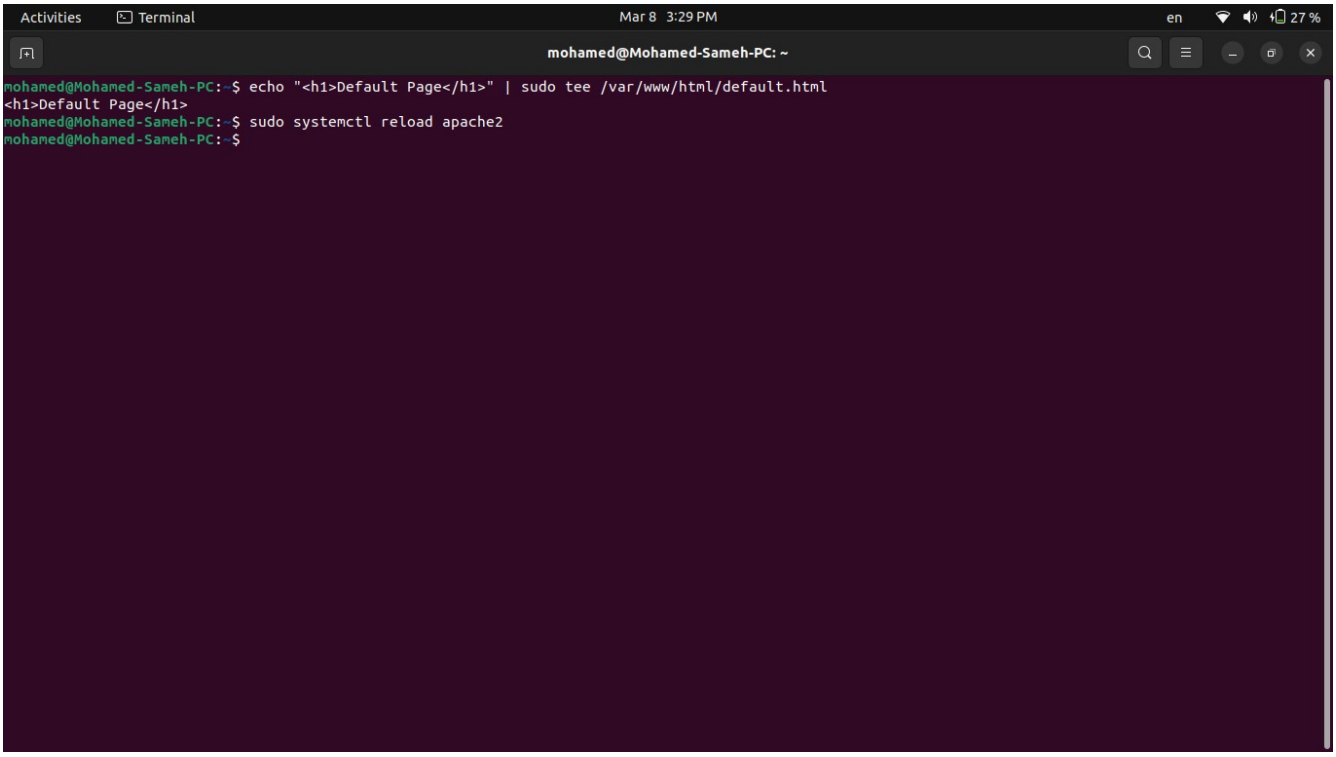
Q5:

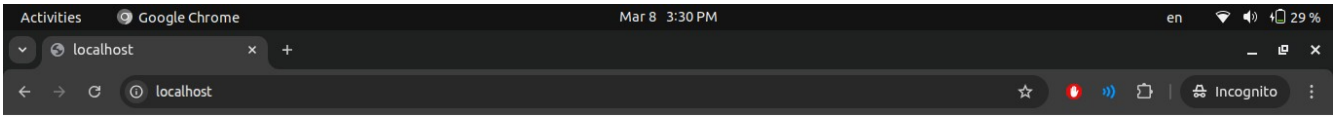

Activities
Terminal
Mar 8 3:20 PM
en
19 %
mohamed@Mohamed-Sameh-PC: ~
mohamed@Mohamed-Sameh-PC:~$ sudo htpasswd /etc/apache2/.htpasswd ahmed
New password:
Re-type new password:
Adding password for user ahmed
mohamed@Mohamed-Sameh-PC:~$ sudo nano /var/www/html/protected/.htaccess
mohamed@Mohamed-Sameh-PC:~$ sudo systemctl reload apache2
mohamed@Mohamed-Sameh-PC:~$

```
Activities   Terminal                    Mar 8  3:20 PM                    en   19 %
mohamed@Mohamed-Sameh-PC: ~
  GNU nano 6.2                          /var/www/html/protected/.htaccess
AuthType Basic
AuthName "Restricted Area"
AuthUserFile /etc/apache2/.htpasswd
Require ahmed

                                        [ Read 4 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^Q Where Was
```

Q6:

```
Activities   Terminal                    Mar 8  3:27 PM                    en   25 %
mohamed@Mohamed-Sameh-PC: ~
  GNU nano 6.2                          /etc/apache2/mods-enabled/dir.conf *
<IfModule mod_dir.c>
        DirectoryIndex default.html index.html index.cgi index.pl index.php index.xhtml index.htm
</IfModule>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^Q Where Was
```





Default Page